Cardiovascular System (CARD) · NPLEX-P1-CARD-16

CV-16 Immune Responses in Cardiovascular Disease

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-16-01-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild atheroma formation as an inflammatory response

A competent response contains

· Atherosclerosis is a chronic inflammatory and healing response of the intima of large and medium arteries to endothelial injury, sustained by LDL retained in the wall.

· Healthy endothelium under laminar shear stress releases nitric oxide and prostacyclin and displays few adhesion molecules.

· High LDL, hypertension, smoking, hyperglycemia, and disturbed flow cause endothelial dysfunction: the intact endothelium becomes more permeable to LDL, makes less nitric oxide, and activates NF-kappa B.

· NF-kappa B induces selectins for leukocyte rolling, VCAM-1 which binds VLA-4, and MCP-1 which draws cells into the intima; M-CSF converts monocytes into macrophages.

· VCAM-1 is selective for mononuclear cells, so early lesions contain monocytes and T cells rather than neutrophils; the T cells are mainly Th1 and their IFN-gamma activates macrophages.

· LDL trapped on intimal proteoglycans is oxidized; oxidized LDL no longer fits the LDL receptor, which SREBP shuts down as cell cholesterol rises.

· Oxidized LDL enters macrophages through scavenger receptors SR-A and CD36, which are not down-regulated by cell cholesterol, so uptake has no brake.

· ACAT esterifies the excess cholesterol into droplets to make foam cells, and efflux to HDL runs through ABCA1 and ABCG1.

· Foam cells form the fatty streak, already present in many children.

· PDGF from macrophages, endothelium, and platelets draws medial smooth muscle cells into the intima, and TGF-beta makes them lay down collagen over a necrotic core of lipid, cholesterol clefts, debris, and tissue factor.

· Plaques form first at the coronary and renal ostia and the carotid and aortic bifurcations, where flow is disturbed rather than laminar.

Distinctions that matter

· Endothelial dysfunction is not endothelial denudation: the layer is still intact but leaky, less protective, and activated.

· The LDL receptor is cholesterol regulated and the scavenger receptor is not, which is why macrophages keep loading until they become foam cells.

Common misconceptions

· Writing that neutrophils start the lesion. VCAM-1 selects monocytes and lymphocytes, which is why early plaques are mononuclear.

· Treating plaque as passive lipid accumulation. Smooth muscle recruitment and collagen synthesis make it a healing response, and it is inflammation that undoes that repair.

Where it is taught: Section C01, opening paragraphs, the recruitment and foam cell paragraphs, and the atheroma flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C01-BD2

Brain dump 2 · integration and application: Why the smaller plaque is the dangerous one

A competent response contains

· Lesion A is the vulnerable plaque and is more likely to cause infarction; lesion B with its thick cap and severe stenosis more typically causes stable angina.

· Macrophage matrix metalloproteinases MMP-1, MMP-8, MMP-13, and MMP-9, along with cathepsins, degrade cap collagen.

· Th1 IFN-gamma blocks collagen synthesis by smooth muscle cells, and those cells undergo apoptosis, so the cap is not repaired.

· Rupture usually begins at the macrophage-rich shoulders of the plaque, exposes tissue factor from the necrotic core, and thrombus forms within minutes.

· A minority of coronary thrombi form over plaque erosion, where endothelium is lost over an intact plaque with no cap tear.

· A thin cap is classically under about 65 micrometers, with a large lipid core, many macrophages and T cells, sparse apoptotic smooth muscle cells, and often only mild to moderate stenosis.

· Rheumatoid arthritis and lupus accelerate atherosclerosis because the same cytokines, IL-1 beta and IL-6, are already circulating.

· An hs-CRP of 4 mg/L is in the higher risk band (below 1 mg/L lower risk, 1 to 3 mg/L average, above 3 mg/L higher) and reflects plaque inflammation.

· The route to CRP is oxidized LDL on Toll-like receptors priming pro-IL-1 beta and NLRP3, cholesterol crystals damaging lysosomes, NLRP3 caspase-1 releasing active IL-1 beta, IL-1 beta inducing IL-6, and IL-6 driving hepatic CRP.

Distinctions that matter

· Degree of stenosis predicts angina; composition and inflammation predict rupture. They are separate questions about the same plaque.

· Rupture exposes the core and its tissue factor, while erosion simply loses the endothelial surface; both end in thrombus.

Common misconceptions

· Assuming the tightest lesion causes the infarct. Many infarcts arise from lesions of only mild to moderate stenosis.

· Reading hs-CRP as a measure of plaque size. It reports inflammatory activity through the IL-1 beta and IL-6 route.

Where it is taught: Section C01, paragraph on the fibrous cap, the stable against vulnerable plaque table, and the stop and predict box of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C01-RT01

Retrieval exercise: Rebuild the atheroma sequence and the inflammasome branch

A competent response contains

· Step 1: endothelial dysfunction from LDL, hypertension, smoking, hyperglycemia, and disturbed flow.

· Step 2: NF-kappa B induces selectins, VCAM-1, and MCP-1.

· Step 3: monocytes and Th1 cells enter the intima; selectins and VCAM-1 made them adhere and MCP-1 drew them in.

· Step 4: retained LDL is oxidized.

· Step 5: scavenger receptor uptake creates foam cells and the fatty streak; SR-A and CD36 are not down-regulated by cell cholesterol, so there is no brake on uptake.

· Step 6: PDGF and TGF-beta build a smooth muscle and collagen cap.

· Step 7: metalloproteinases and IFN-gamma thin the cap.

· Step 8: rupture exposes tissue factor and thrombus forms.

· Inflammasome branch: oxidized LDL on Toll-like receptors, macrophage makes pro-IL-1 beta and NLRP3, cholesterol crystals damage lysosomes, NLRP3 inflammasome assembles, caspase-1 cleaves pro-IL-1 beta, active IL-1 beta is released, IL-1 beta induces IL-6, and the liver produces CRP.

· hs-CRP bands: below 1 mg/L lower risk, 1 to 3 mg/L average, above 3 mg/L higher risk.

Distinctions that matter

· Priming and activation are two separate signals: Toll-like receptors supply the pro-form and the sensor, and crystals supply the trigger.

· IL-1 beta acts locally on the plaque while IL-6 carries the signal to the liver, which is why a blood test reflects wall inflammation.

Common misconceptions

· Putting cap formation before foam cells. Smooth muscle recruitment is the repair response to an established lipid lesion.

· Describing CRP as made by the plaque. It is a hepatic acute phase protein induced by IL-6.

Where it is taught: Section C01, atheroma flow figure and the IL-1 beta and CRP paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-BD1

Brain dump 1 · foundational retrieval: Rheumatic fever from the throat to the valve

A competent response contains

· Sequence: group A streptococcal pharyngitis in a susceptible child; antibodies and T cells form against M protein and group A carbohydrate; about 2 to 3 weeks later they cross-react with myosin and valve proteins; activated valve endothelium admits CD4 T cells; pancarditis with Aschoff bodies and verrucae on the lines of closure; repeated attacks heal by fibrosis into chronic mitral stenosis.

· The organism is Streptococcus pyogenes, the infection is pharyngitis rather than skin, and the usual age is about 5 to 15 years.

· The M protein is an alpha-helical coiled coil resembling cardiac myosin and tropomyosin, and antibody to the N-acetylglucosamine of the group A carbohydrate cross-reacts with valve laminin.

· Cross-reactive antibody activates valve endothelium and raises VCAM-1, letting CD4 T cells enter and sustain inflammation with IFN-gamma and IL-17.

· The disease is classed as type II hypersensitivity, but chronic valve damage is driven largely by T cells.

· Risk is higher with certain HLA class II alleles and rheumatogenic strains, and the organism is absent from the heart lesions.

· The Aschoff body is fibrinoid collagen surrounded by T cells, plasma cells, and Anitschkow cells, activated macrophages with a wavy ribbon of chromatin.

· Aschoff bodies occur in all three layers, so the carditis is a pancarditis: fibrinous pericarditis resolves, myocarditis is the main cause of acute death, and valvulitis leaves lasting damage.

· Acute verrucae are about 1 to 2 mm along the lines of closure, mitral far more often than aortic.

· Repeated attacks give leaflet thickening, commissural fusion, and short fused chordae, producing fish-mouth mitral stenosis.

Distinctions that matter

· The latent period separates rheumatic carditis from infective endocarditis: here the organism has gone before the heart is injured.

· Mimicry follows throat infection only, while poststreptococcal glomerulonephritis follows throat or skin infection because it is an immune complex disease.

Common misconceptions

· Looking for bacteria in the valve lesion. The lesions are sterile and the injury is immune.

· Calling acute rheumatic valvulitis a stenosis. The acute lesion usually causes mitral regurgitation, and stenosis is the scar of repeated attacks.

Where it is taught: Section C02, opening paragraphs, the mimicry flow figure, and the pathology paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-BD2

Brain dump 2 · integration and application: Work a suspected case from timing and serology

A competent response contains

· Major criteria present: carditis (new murmur) and polyarthritis, which are migratory and in large joints.

· Minor criteria present: fever of 38.7 degrees C (101.7 degrees F), which meets the low-risk threshold of 38.5 degrees C (101.3 degrees F) or higher, and an ESR of 72 mm/h, which meets the threshold of 60 mm/h or higher.

· Two major criteria with evidence of recent streptococcal infection meet the requirement for an initial attack.

· The prolonged PR interval cannot be counted here, because carditis is already serving as a major criterion.

· Serology: antistreptolysin O rises after about one week and peaks at about 3 to 6 weeks; anti-DNase B peaks later and persists longer, which helps when chorea appears late. A rising titer is stronger evidence than a single value.

· The joints recover because the arthritis is a transient exudative synovitis that heals without deformity; the valve scars because valve inflammation heals by fibrosis.

· The child with involuntary writhing movements months later has Sydenham chorea, from antibodies cross-reacting with basal ganglia neurons, and it can support the diagnosis alone after titers have fallen.

· The child with skin sores, dark urine, and red cell casts has poststreptococcal glomerulonephritis, an immune complex disease, which is why it can follow either pharyngitis or impetigo.

· The site of infection decides the answer because mimicry of cardiac proteins follows throat infection, whereas immune complexes can form after infection at either site.

· Recurrences after new streptococcal infections are common and each adds valve scarring.

Distinctions that matter

· Higher risk populations use lower thresholds: fever 38.0 degrees C (100.4 degrees F), ESR 30 mm/h, and monoarthritis or polyarthralgia counted as major.

· A negative throat culture does not argue against the diagnosis, because the disease appears weeks after the organism has cleared.

Common misconceptions

· Counting the prolonged PR interval alongside carditis. It is excluded when carditis is a major criterion so that one process is not counted twice.

· Calling late chorea unrelated because titers have fallen. Anti-DNase B persists longer and chorea can carry the diagnosis by itself.

Where it is taught: Section C02, Jones criteria table, serology paragraph, and the stop and predict box of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-BD1

Brain dump 1 · foundational retrieval: Organize the vasculitides by immune effector

A competent response contains

· Immune complex group: polyarteritis nodosa, IgA vasculitis, and cryoglobulinemic vasculitis.

· ANCA group: granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis.

· Granulomatous T-cell group: giant cell arteritis and Takayasu arteritis.

· Immune complexes form in slight antigen excess, escape clearance because they are small, and deposit where flow is turbulent or filtered, as in glomeruli and dermal venules.

· They activate the classical pathway, C5a recruits neutrophils, and neutrophil enzymes and oxidants cause fibrinoid necrosis; complement may fall and immunofluorescence shows granular immunoglobulin and complement.

· Nuclear dust from dying neutrophils gives leukocytoclastic vasculitis, and in dermal venules the lesion is palpable purpura: raised, nonblanching lesions on dependent skin.

· ANCA are IgG against PR3, giving cytoplasmic staining, or MPO, giving perinuclear staining because ethanol fixation lets cationic MPO move to the nucleus.

· Infection, TNF, or C5a primes neutrophils and moves these enzymes to the surface; ANCA bind by Fab and engage Fc-gamma receptors on the same cell.

· The neutrophil then releases oxidants, proteases, and extracellular traps onto endothelium and activates the alternative pathway for more C5a; because the neutrophil delivers the injury, lesions are pauci-immune and complement is normal.

· In giant cell and Takayasu arteritis, adventitial dendritic cells activate CD4 T cells, IFN-gamma recruits macrophages and giant cells into the media around a fragmented internal elastic lamina, intimal hyperplasia narrows the lumen, there are no deposits, and IL-6 drives a very high ESR and CRP.

Distinctions that matter

· Granular deposits mean immune complexes, few or no deposits mean ANCA or T-cell disease, and linear deposits mean antibody against fixed type IV collagen in anti-GBM disease.

· Immune complex disease is often complement consuming while ANCA disease uses the alternative pathway locally and leaves serum complement normal.

Common misconceptions

· Grouping vasculitis by vessel size when the question is about serology. Size predicts the organs involved; mechanism predicts the complement level and the immunofluorescence.

· Treating pauci-immune as meaning no antibody involved. ANCA is the antibody; it simply acts on the neutrophil instead of depositing in the wall.

Where it is taught: Section C03, immune complex paragraphs, the ANCA and granulomatous paragraphs, and the vasculitis table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-BD2

Brain dump 2 · integration and application: Name the vasculitis from complement and serology

A competent response contains

· Patient 1 has mixed cryoglobulinemic vasculitis: hepatitis C drives clonal B cells to make IgM rheumatoid factor that binds IgG, and these complexes consume the classical pathway, so C4 is very low; the complexes precipitate below 37 degrees C (98.6 degrees F).

· Patient 2 has microscopic polyangiitis: MPO-ANCA with pulmonary hemorrhage and crescentic glomerulonephritis and no granulomas; a pauci-immune biopsy settles it.

· Patient 3 has IgA vasculitis: complexes of galactose-deficient IgA1 deposit in skin, gut, joints, and mesangium after a respiratory infection, and mesangial IgA settles it.

· Patient 4 has polyarteritis nodosa: medium and small muscular arteries, in some cases through hepatitis B antigen-antibody complexes, ANCA negative, with lungs spared.

· C4 is very low in patient 1 because IgM-IgG complexes drive the classical pathway hard.

· Complement stays normal in patient 3 because IgA uses the alternative and lectin pathways rather than the classical pathway.

· Patient 2 has no deposits because the primed neutrophil, armed by ANCA, delivers the injury directly rather than complexes settling in the wall.

· Patient 4 has renal artery lesions with infarcts and renin-dependent hypertension but no glomerulonephritis, because the disease is in muscular arteries rather than in glomerular capillaries; mesenteric lesions cause postprandial pain and vasa nervorum lesions cause mononeuritis multiplex.

· The 74-year-old has giant cell arteritis and the 29-year-old has Takayasu arteritis; both are type IV granulomatous T-cell disease of large arteries with IL-6 driving a high ESR.

· Giant cell arteritis involves cranial carotid branches and the vertebral arteries, giving temporal headache, jaw claudication, and vision loss from posterior ciliary artery occlusion; Takayasu arteritis involves the aorta and its main branches in patients under 40, often women, giving weak arm pulses and an arm blood pressure difference.

Distinctions that matter

· Low C4 with near normal C3 points to classical pathway consumption by IgM-IgG cryoglobulins, whereas IgA vasculitis spares C4 even though complexes are deposited.

· Segmental lesions of different ages with microaneurysms and no lung involvement separate polyarteritis nodosa from the ANCA diseases.

Common misconceptions

· Expecting glomerulonephritis in polyarteritis nodosa. The lesions are in muscular arteries, so the kidney shows infarcts and renin-dependent hypertension instead.

· Reading normal complement as evidence against immune complex disease. IgA vasculitis has deposits with normal complement.

Where it is taught: Section C03, immune complex and ANCA paragraphs, the vasculitis table, and the worked example on using complement of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-CP01

Comparison table: Immune complex, ANCA, and granulomatous vasculitis side by side

A competent response contains

· Type and effector: immune complex disease is type III with neutrophils recruited by C5a; ANCA disease is antibody directed at primed neutrophils, which deliver the injury; large vessel arteritis is type IV with CD4 T cells, macrophages, and giant cells.

· Immunofluorescence: granular immunoglobulin and complement in immune complex disease; few or no deposits, described as pauci-immune, in ANCA disease; no deposits in granulomatous arteritis.

· Complement: may fall in immune complex disease because the classical pathway is consumed, with very low C4 in cryoglobulinemia and normal complement in IgA vasculitis which uses the alternative and lectin pathways; normal in ANCA disease, where the alternative pathway is activated locally; normal in granulomatous arteritis.

· Markers: hepatitis B or C serology, rheumatoid factor and cryoglobulins, and serum complement for the complex diseases; PR3-ANCA and MPO-ANCA for the ANCA diseases; a very high ESR and CRP driven by IL-6 for the granulomatous diseases.

· Vessels and tissues: small vessels and dermal venules and glomeruli, with medium muscular arteries in polyarteritis nodosa; small vessels of lung and kidney with upper airway granulomas in granulomatosis with polyangiitis; the aorta and its large branches, cranial carotid branches, and vertebral arteries in the granulomatous group.

· Examples: polyarteritis nodosa, IgA vasculitis, and cryoglobulinemic vasculitis; granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis; giant cell arteritis and Takayasu arteritis.

· Findings: palpable purpura with leukocytoclastic change, arthralgia, abdominal pain, and nephritis in complex disease; crescentic glomerulonephritis with pulmonary hemorrhage, and sinus and nasal granulomas when PR3-ANCA is present; headache, jaw claudication, vision loss, polymyalgia rheumatica, weak arm pulses, and an arm blood pressure difference in the granulomatous group, with the skin and kidney spared.

Distinctions that matter

· The immunofluorescence pattern is the fastest discriminator: granular means complexes, pauci-immune means ANCA, and no deposits with giant cells means T-cell disease.

· Within the ANCA group, PR3 with granulomas of the sinuses and lungs means granulomatosis with polyangiitis, MPO without granulomas means microscopic polyangiitis, and asthma with eosinophilia driven by IL-5 means eosinophilic granulomatosis with polyangiitis.

Common misconceptions

· Expecting a low complement in every vasculitis with deposits. IgA vasculitis has normal complement because of the pathway it uses.

· Expecting skin purpura and casts in giant cell arteritis. It is a large artery disease, so the findings follow the branches involved.

Where it is taught: Section C03, all paragraphs and the vasculitis by immune mechanism table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-BD1

Brain dump 1 · foundational retrieval: Three immune routes into the heart

A competent response contains

· Dressler syndrome: necrosis of a transmural infarct releases sequestered myocardial proteins, antibodies form against them, and immune complexes are thought to deposit on pericardium and pleura; it appears weeks to months later, and the delay is the time to mount that response.

· Early post-infarct pericarditis within about 1 to 3 days is a local fibrinous reaction to necrosis rather than an autoimmune process.

· Dressler findings: fever, pleuritic pain relieved by sitting forward, a friction rub, pleural and pericardial effusions, and high ESR and CRP, with diffuse ST elevation and PR depression rather than the regional changes of the infarct.

· Post-pericardiotomy syndrome has the same mechanism.

· Libman-Sacks endocarditis: immune complex and complement injury of valve endothelium in lupus, aided by antiphospholipid antibodies that promote thrombosis, gives sterile verrucous vegetations of fibrin, platelets, and mononuclear cells, sometimes with hematoxylin bodies.

· Those vegetations can sit on either leaflet surface, the mitral valve most often, heal by fibrosis that causes regurgitation, and can embolize fragments to the brain.

· Congenital heart block: maternal IgG against Ro (SSA), often with La (SSB), crosses the placenta on the neonatal Fc receptor from the second trimester; the mother may have lupus, Sjogren syndrome, or no symptoms.

· Antibody binds Ro and La exposed on apoptotic fetal myocytes, macrophages release TNF and TGF-beta, and fibrosis and calcification replace the AV node at about 18 to 24 weeks.

· Anti-Ro also inhibits L-type and T-type Ca2+ channels, and since AV nodal cells depolarize by Ca2+ current, conduction slows even before scarring.

· The three routes differ: antigens exposed by injury in Dressler syndrome, systemic autoimmunity in Libman-Sacks endocarditis, and maternal antibody crossing the placenta in congenital heart block.

Distinctions that matter

· Timing separates the two post-infarct pericarditides: days means a local reaction to necrosis, weeks to months means an autoimmune response.

· Diffuse ST elevation with PR depression is pericardial and involves the whole surface, while the infarct gives changes confined to a lead group.

Common misconceptions

· Treating Libman-Sacks vegetations as infected. They are sterile, and their position on either leaflet surface is part of what identifies them.

· Expecting the mother of an affected newborn to be unwell. She may have no symptoms and still carry anti-Ro.

Where it is taught: Section C04, Dressler and Libman-Sacks paragraphs, the vegetation table, and the congenital heart block paragraphs of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-BD2

Brain dump 2 · integration and application: Explain the newborn and sort four sets of vegetations

A competent response contains

· The atria beat normally at 142 per minute while the ventricles follow a slow escape rhythm at 50 per minute with no relation between them: complete heart block with AV dissociation, below the normal fetal rate of about 110 to 160 bpm.

· The immediate effect is inhibition of L-type and T-type Ca2+ channels by anti-Ro, which slows conduction in a node that depolarizes by Ca2+ current; the permanent effect is fibrosis and calcification of the node.

· Macrophages responding to antibody bound on apoptotic fetal myocytes release TNF and TGF-beta, which drive that fibrosis.

· The node is injured at about 18 to 24 weeks of gestation.

· By 8 months the rash, cytopenias, and hepatitis of neonatal lupus have resolved as maternal IgG is cleared over about 6 to 8 months, but the fibrosed AV node cannot regenerate, so the block is permanent.

· Risk is about 1 to 2% per anti-Ro positive pregnancy and about 15 to 20% after an affected child.

· Rheumatic vegetations: molecular mimicry, small at 1 to 2 mm, on the lines of closure, no organisms, and fibrosis with stenosis after repeated attacks.

· Libman-Sacks vegetations: immune complexes with antiphospholipid antibodies, small to medium at 1 to 4 mm, on either or both leaflet surfaces, no organisms, healing to fibrosis and regurgitation.

· Infective vegetations: bacteria or fungi, large and friable, on the upstream surface of the closure line, organisms present, with destruction, perforation, and abscess.

· Nonbacterial thrombotic vegetations: a hypercoagulable state, often cancer, small to medium, on the lines of closure, no organisms, little valve damage but bland emboli.

Distinctions that matter

· Antibody effects that depend on continued antibody presence resolve when IgG is catabolized; effects that have already become scar do not.

· Only infective vegetations contain organisms, and position on the leaflet separates the three sterile types.

Common misconceptions

· Expecting the bradycardia to lift with the rash. The channel block is reversible but the nodal scar is not.

· Calling every sterile vegetation Libman-Sacks. Rheumatic and nonbacterial thrombotic vegetations sit on the lines of closure, while Libman-Sacks lesions can sit on either surface.

Where it is taught: Section C04, congenital heart block flow figure and final paragraph, and the vegetation comparison table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C04-RT01

Retrieval exercise: Rebuild the pathway to congenital heart block

A competent response contains

· Step 1: the mother has IgG anti-Ro, often with anti-La.

· Step 2: FcRn, the neonatal Fc receptor, carries the IgG across the placenta from the second trimester.

· Step 3: antibody binds Ro and La exposed on apoptotic fetal myocytes and also inhibits L-type and T-type Ca2+ channels.

· Step 4: macrophages release TNF and TGF-beta.

· Step 5: fibrosis and calcification replace the AV node at about 18 to 24 weeks.

· Step 6: permanent complete heart block with a slow escape rhythm, well below the normal fetal rate of about 110 to 160 bpm.

· The fibrosis step makes the block permanent, because nodal conduction tissue cannot regenerate once it has been replaced by scar and calcium.

Distinctions that matter

· Antibody binding and channel inhibition act while the antibody is present; fibrosis outlives it.

· AV nodal cells depend on Ca2+ current for their upstroke, which is why an antibody that blocks Ca2+ channels affects this site in particular.

Common misconceptions

· Treating the antibody as directly cytotoxic to nodal tissue. The lasting damage comes from the macrophage response and the fibrosis it drives.

Where it is taught: Section C04, pathogenesis of congenital heart block flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-BD1

Brain dump 1 · foundational retrieval: Build the rejection table from memory

A competent response contains

· Hyperacute: minutes to hours, preformed anti-ABO IgM or anti-HLA IgG, type II, with small vessel thrombosis and necrosis.

· Acute cellular: weeks to months and mostly in the first year, CD8 and Th1 T cells, type IV, with perivascular and interstitial lymphocytes and myocyte damage.

· Antibody-mediated: days to years, donor-specific anti-HLA antibody, type II, with endothelial swelling, intravascular macrophages, and capillary C4d.

· Allograft vasculopathy: months to years, T cells, antibody, and nonimmune injury, mixed mechanism, with diffuse concentric intimal thickening.

· Direct allorecognition: recipient T cells see intact donor HLA on donor dendritic cells, the passenger leukocytes carried in the graft; alloreactive T cells are numerous, so this drives the strong early response.

· Indirect allorecognition: recipient antigen-presenting cells display processed donor HLA peptides on recipient class II molecules, so it persists after passenger leukocytes are gone and gives B cells the CD4 help to make donor-specific antibody, which is why it matters in antibody-mediated and chronic rejection.

· Preformed antibody binds endothelium, activates the classical pathway, and small vessels thrombose at once.

· In cellular rejection CD8 T cells kill myocytes with perforin, granzymes, and Fas ligand, and Th1 IFN-gamma activates macrophages and raises graft HLA expression; it is often clinically silent.

· C4d marks the capillaries because it binds covalently to tissue where C4 was cleaved, so it remains as a footprint of classical pathway activation.

· Allograft vasculopathy gives diffuse concentric intimal thickening of epicardial and intramyocardial arteries with an intact internal elastic lamina and little lipid or calcium.

· Hyperacute rejection is now rare because of ABO matching and crossmatch screening for anti-HLA antibody.

Distinctions that matter

· Timing follows the effector: preformed antibody needs no time, naive T cells need costimulation, IL-2, and clonal expansion, and vessel remodeling takes years.

· Cellular rejection is interstitial and lymphocytic while antibody-mediated rejection is microvascular and complement marked, which is why C4d is the discriminator.

Common misconceptions

· Reading antibody-mediated rejection as a late process only. It occurs from days to years, whenever donor-specific antibody appears.

· Assuming higher biopsy grade always means more illness. Widespread microvascular injury compromises hemodynamics more than cellular rejection of similar grade.

Where it is taught: Section C05, allorecognition paragraph, the four types, and the rejection table of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-BD2

Brain dump 2 · integration and application: Three recipients and a denervated heart

A competent response contains

· Recipient 1 has acute cellular rejection: CD8 and Th1 T cells, and 3 weeks fits the time naive T cells need for costimulation, IL-2, and clonal expansion.

· Recipient 2 has antibody-mediated rejection: donor-specific anti-HLA antibody on capillary endothelium with classical pathway activation, which C4d records; indirect allorecognition supplied the CD4 help for the antibody to appear after transplantation.

· Recipient 3 has cardiac allograft vasculopathy, the chronic form, which takes months to years of remodeling.

· Recipient 2 is more unwell because microvascular injury is widespread and compromises perfusion of the whole graft, more than cellular rejection of similar grade.

· Allograft vasculopathy differs from atherosclerosis by being diffuse and concentric rather than focal and eccentric, by leaving the internal elastic lamina intact, and by containing little lipid or calcium.

· Immune drivers are indirect allorecognition, IFN-gamma, and donor-specific antibody; nonimmune drivers are ischemia-reperfusion injury, cytomegalovirus, hyperlipidemia, and hypertension; endothelium releases PDGF and TGF-beta that drive smooth muscle into the intima.

· Angiography understates it because diffuse narrowing leaves no normal reference segment to compare against.

· All three have a resting rate near the intrinsic SA nodal rate of about 90 to 110 bpm, because the graft is denervated and has no vagal tone.

· Exercise raises the rate slowly, through circulating catecholamines rather than sympathetic nerves.

· Recipient 3 has no angina because the graft has no sensory afferents, so vasculopathy presents as heart failure, arrhythmia, silent infarction, or sudden death.

· Thrombosis within minutes of reperfusion is hyperacute rejection, from preformed anti-ABO IgM or anti-HLA IgG.

Distinctions that matter

· C4d with new donor-specific antibody separates antibody-mediated from cellular rejection even when both show infiltrating cells.

· Denervation explains both the high resting rate and the absence of angina, so the same fact answers two different questions.

Common misconceptions

· Expecting chest pain to warn of graft coronary disease. The graft is denervated, so the presentation is painless.

· Reading a normal coronary angiogram as excluding allograft vasculopathy. The disease is diffuse, so there is no narrower segment to see.

Where it is taught: Section C05, the four types, the denervation paragraph, the rejection table, and the worked example on matching biopsies of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-BD1

Brain dump 1 · foundational retrieval: Viral myocarditis in three phases

A competent response contains

· Myocarditis is myocardial inflammation with myocyte injury not explained by ischemia.

· Phase 1, days: coxsackievirus B enters myocytes through the coxsackievirus and adenovirus receptor with CD55, viral protease 2A cleaves dystrophin and kills myocytes, and type I interferons, NK cells, and

macrophages limit spread. Adenovirus, parvovirus B19, HHV-6, influenza, and SARS-CoV-2 are other causes.

· Phase 2, from about 1 to 2 weeks: CD8 T cells kill infected myocytes and CD4 T cells release IFN-gamma and TNF; this clears virus but causes most of the muscle loss, and released myosin with molecular mimicry can trigger autoimmunity.

· Phase 3: most patients recover, and a minority develop dilated cardiomyopathy from persistent virus or from autoimmunity, with fibrosis and remodeling over months to years.

· Diagnosis: the Dallas criteria require an inflammatory infiltrate with nonischemic myocyte damage on biopsy, but biopsy can miss patchy disease, so MRI is used.

· On MRI, late gadolinium enhancement is subepicardial or midwall, unlike the subendocardial start of infarction.

· Necrosis raises troponin.

· Myocardial and pericardial inflammation cause chest pain and ST changes outside a single coronary territory.

· Muscle loss causes heart failure, and fulminant disease can recover substantially because many myocytes are dysfunctional rather than dead.

· Patchy scar supports reentry, so ventricular arrhythmias and sudden death occur, sometimes in young athletes.

· A viral prodrome often precedes the cardiac symptoms.

Distinctions that matter

· Direct viral killing belongs to the first days; the immune phase from 1 to 2 weeks accounts for most myocyte loss.

· Enhancement that starts subepicardially is inflammatory, while infarction starts subendocardially where perfusion is last and least.

Common misconceptions

· Assuming the virus does most of the damage. The CD8 response that clears it causes most of the muscle loss.

· Reading ST changes with a raised troponin as an infarct. The changes here cross coronary territories and the arteries are normal.

Where it is taught: Section C06, opening paragraph, the three phases, the flow figure, and the clinical paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-BD2

Brain dump 2 · integration and application: Separate four patients with myocarditis

A competent response contains

· Patient 1 has viral myocarditis in the immune phase: CD8 T cells with perforin, granzymes, and Fas ligand, with CD4 cells releasing IFN-gamma and TNF; biopsy shows lymphocytes with myocyte necrosis.

· Patient 2 has hypersensitivity myocarditis, a dose-independent type IV drug reaction in which Th2 cytokines such as IL-5 recruit eosinophils; biopsy shows T cells and eosinophils with limited necrosis, and sulfonamides, beta-lactams, methyldopa, and clozapine are examples.

· Patient 3 has hypereosinophilic myocarditis, from hypereosinophilic syndrome or eosinophilic granulomatosis with polyangiitis; biopsy shows eosinophil granule injury with necrosis and mural thrombus.

· Patient 4 has myocarditis after mRNA COVID-19 vaccination, which clusters in young males after the second dose, begins within days, and is usually mild; the mechanism is not established and MRI shows subepicardial enhancement.

· Patient 3 is the one who ends with restrictive physiology: eosinophil major basic protein and cationic protein kill myocytes and injure endocardium, mural thrombus organizes, and endomyocardial fibrosis, also called Loffler endocarditis, stiffens the ventricle.

· Patient 2 should subside once the drug is stopped, because the reaction depends on continued exposure to the drug, and it is dose independent.

· The fifth patient has giant cell myocarditis: T cells with giant cells and wide necrosis and no granulomas, associated with autoimmunity and thymoma, with rapid heart failure, arrhythmias, and a poor prognosis.

· Eosinophilia with asthma and a neuropathy also raises eosinophilic granulomatosis with polyangiitis, where IL-5 drives the eosinophils and MPO-ANCA is found in about 40%.

· The effector cell on histology is what predicts the course in each form.

Distinctions that matter

· Drug hypersensitivity and hypereosinophilic disease both bring eosinophils, but one subsides when exposure ends and the other scars the endocardium.

· Giant cell myocarditis has giant cells and wide necrosis without granulomas, which separates it from granulomatous myocarditis.

Common misconceptions

· Expecting a drug reaction in the heart to be dose related. Hypersensitivity myocarditis is a type IV reaction and is dose independent.

· Grouping all eosinophilic myocardial disease together. The granule proteins of a sustained hypereosinophilia are what produce endomyocardial fibrosis.

Where it is taught: Section C06, other forms paragraph and the table comparing forms of myocarditis of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-PR01

Prediction exercise: Predict what the T-cell phase does to the myocardium

A competent response contains

· Virus in the myocardium falls, because CD8 T cells kill infected myocytes and clear the infection.

· Surviving myocytes fall further than in the first days, because immune killing accounts for most of the muscle loss.

· Serum troponin rises, because ongoing myocyte necrosis releases it.

· Dystrophin and the myocyte membrane are already compromised, because viral protease 2A cleaved dystrophin in the first phase and the cell cannot maintain its membrane.

· Exposure to cardiac myosin rises and so does the chance of autoimmunity, because myosin released from dying myocytes can cross-react by molecular mimicry.

· In the minority who do not recover, the chamber dilates and the wall thins as lost muscle is replaced by fibrosis and the ventricle remodels into a dilated cardiomyopathy.

· The risk of reentrant ventricular arrhythmia rises, because patchy fibrosis between surviving muscle creates slow conduction and the circuit conditions reentry needs.

Distinctions that matter

· Clearing the virus and preserving the muscle are not the same goal, which is why the phase that cures the infection is the phase that costs the most myocytes.

· Dilatation reflects remodeling after muscle loss, not continuing infection.

Common misconceptions

· Expecting troponin to fall once the virus is being cleared. Necrosis is greatest while the T-cell response is at its peak.

· Assuming every patient progresses to dilated cardiomyopathy. Most recover, and progression needs persistent virus or autoimmunity.

Where it is taught: Section C06, the three phases, the flow figure from viral infection to dilated cardiomyopathy, and the clinical paragraph of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-BD1

Brain dump 1 · foundational retrieval: HHV-8 latency and the cytokines of myxoma

A competent response contains

· Kaposi sarcoma is a low-grade vascular tumor caused by human herpesvirus 8, a gamma-herpesvirus spread mainly by saliva and also sexually.

· HHV-8 latently infects spindle cells of lymphatic endothelial type.

· LANA keeps the viral episome in the cell and inhibits p53 and Rb.

· v-cyclin pairs with CDK6 to drive the cell from G1 into S phase.

· v-FLIP activates NF-kappa B and blocks apoptosis.

· A viral G protein-coupled receptor induces VEGF, and viral IL-6 signals through gp130 without needing the IL-6 receptor alpha chain.

· Histology: spindle cells with slit-like vascular spaces, extravasated red cells, and hemosiderin, progressing from patches to plaques to nodules in skin, oral mucosa, gut, lymph nodes, and lungs; LANA staining confirms HHV-8.

· The same virus causes primary effusion lymphoma and multicentric Castleman disease.

· Myxoma is the most common primary cardiac tumor in adults, rhabdomyoma is the most common in children, and metastases outnumber both.

· About 90% of myxomas arise in the atria and roughly 80% in the left atrium, as a pedunculated gelatinous mass on the septum near the fossa ovalis, with stellate cells in a glycosaminoglycan-rich matrix.

· Most are sporadic in middle-aged women; familial cases occur in Carney complex from PRKAR1A mutation, and are younger and multiple with spotty skin pigmentation and endocrine tumors.

Distinctions that matter

· Latent genes keep the infected cell alive and dividing while lytic genes supply paracrine growth and angiogenic signals to neighbouring cells.

· Viral IL-6 bypasses the IL-6 receptor alpha chain, so it can act on cells that would not respond to the host cytokine.

Common misconceptions

· Calling Kaposi sarcoma a tumor of blood vessel endothelium. The spindle cell resembles lymphatic endothelium.

· Treating a cardiac mass as most likely primary. Metastases to the heart are more common than either primary tumor.

Where it is taught: Section C07, Kaposi sarcoma paragraph and the atrial myxoma paragraphs of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-BD2

Brain dump 2 · integration and application: Failed surveillance, and one cytokine imitating three diseases

A competent response contains

· CD8 T cells and NK cells normally hold HHV-8 in check, and the CD8 response depends on CD4 help.

· The patient with untreated HIV and a CD4 count of 90 cells per microlitre has AIDS-associated Kaposi sarcoma, the most aggressive form and AIDS-defining: CD4 loss removes the help the CD8 response needs, and HIV Tat further stimulates the infected cells.

· The transplant recipient on a calcineurin inhibitor has lost T-cell activation to the drug, so latent virus is no longer controlled.

· The 68-year-old man of Mediterranean descent has classic Kaposi sarcoma, which is indolent on the lower legs; endemic African disease can involve lymph nodes in children.

· Early lesions depend on viral and cytokine signals, so restoring CD4 counts or reducing immunosuppression often makes them regress, after a possible transient flare from immune reconstitution, while advanced monoclonal lesions may not respond.

· The HIV protein that adds to the proliferation is Tat.

· The woman has an atrial myxoma, and IL-6 secreted by the myxoma cells explains the whole systemic picture.

· IL-6 to fever: COX-2 and PGE2 in the hypothalamus raise the thermal set point. IL-6 to the liver: CRP and fibrinogen raise the ESR, and hepcidin produces the anemia of inflammation. IL-6 on B cells: polyclonal hypergammaglobulinemia. IL-6 driven catabolism and anorexia: weight loss and fatigue.

· The positional symptoms come from the pedunculated mass prolapsing into the mitral orifice in diastole, which obstructs filling and causes dyspnea or syncope with a tumor plop in early diastole.

· The mass is friable, so fragments embolize systemically.

· All the systemic findings resolve after the mass is removed, because the source of IL-6 is gone.

Distinctions that matter

· Immune surveillance failure explains why the same virus produces indolent disease in one host and aggressive disease in another.

· A tumor can produce fever, anemia, and a high ESR through a cytokine alone, with no infection and no autoimmunity.

Common misconceptions

· Expecting all Kaposi lesions to regress when immunity returns. Advanced monoclonal lesions may not, and a transient flare can come first.

· Investigating the myxoma picture as infection or vasculitis without looking at the heart. Sterile cultures with negative autoantibodies and normal complement point away from both.

Where it is taught: Section C07, immune surveillance paragraph and the myxoma paragraphs with the IL-6 flow figure of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C07-CA01

Clinical application: Fever, weight loss, and a high ESR in a woman with positional breathlessness

A competent response contains

· Recognize: a left atrial myxoma, pedunculated and attached to the interatrial septum near the fossa ovalis, which is the characteristic site; about 90% of myxomas are atrial and roughly 80% left atrial.

· Explain: myxoma cells secrete IL-6. In the hypothalamus COX-2 and PGE2 raise the set point and cause fever; in the liver CRP and fibrinogen raise the ESR and hepcidin produces the anemia of inflammation, which fits a normal to raised ferritin with a normocytic anemia; on B cells IL-6 drives polyclonal hypergammaglobulinemia; IL-6 driven catabolism and anorexia cause the weight loss and fatigue.

· Differentiate: infective endocarditis would have organisms and positive blood cultures, and its vegetations sit on valve leaflets rather than on the septum; a systemic vasculitis would be expected to show ANCA or complement consumption with purpura, casts, or a neuropathy; lupus would show antinuclear antibody and low complement and would give sterile valve vegetations on the leaflets rather than a pedunculated septal mass.

· Investigate: look at the mass itself for a pedicle arising near the fossa ovalis, mobility with diastolic prolapse toward the mitral orifice, and a gelatinous friable surface; histology shows stellate cells in a glycosaminoglycan-rich matrix; ask about younger age, multiple tumors, spotty skin pigmentation, and endocrine tumors for Carney complex with PRKAR1A mutation.

· Decide: the positional breathlessness comes from the mobile mass prolapsing into the mitral orifice in diastole and obstructing ventricular filling, with the early diastolic sound being a tumor plop; the arm weakness episodes are systemic emboli, permitted by the friable surface of the mass.

· Justify: a benign tumor mimics chronic infection or malignancy because one cytokine reproduces the acute phase response, and all the systemic findings, the fever, anemia, high ESR, and raised immunoglobulins, resolve after the mass is removed because the source of IL-6 is gone.

Distinctions that matter

· Position settles the lesion: a pedunculated mass on the interatrial septum is a myxoma, while vegetations sit on valve leaflets.

· Anemia of inflammation through hepcidin is not iron deficiency, which is why the ferritin is not low.

Common misconceptions

· Reading fever with a high ESR and raised immunoglobulins as proof of infection or autoimmunity. A cytokine-secreting tumor produces the same profile.

· Expecting obstruction from a cardiac mass to be constant. A mobile pedunculated tumor obstructs only when it falls into the orifice, which is why the symptoms are positional.

Where it is taught: Section C07, atrial myxoma paragraphs and the flow figure on how a myxoma causes constitutional symptoms of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C02-Q01

Board-style question: Carditis three weeks after a sore throat

Correct answer: B. Antibodies and T cells raised against streptococcal M protein and group A carbohydrate that also recognize cardiac myosin and valve proteins

Why it is right: Antibodies and T cells raised against M protein and the group A carbohydrate cross-react with cardiac myosin and valve laminin; cross-reactive antibody activates valve endothelium and raises VCAM-1, letting CD4 T cells enter and sustain the carditis with IFN-gamma and IL-17.

A is wrong: Bacterial invasion with endocardial destruction describes infective endocarditis, which produces large friable vegetations with organisms present, whereas rheumatic lesions are sterile and the organism has already cleared.

C is wrong: Immune complex deposition with classical pathway activation is the mechanism of poststreptococcal glomerulonephritis, which is why that disease can follow skin infection as well as pharyngitis.

D is wrong: Antineutrophil cytoplasmic antibodies act on primed neutrophils in the ANCA vasculitides and produce pauci-immune small vessel injury, not post-streptococcal carditis.

A competent response contains

· Migratory large joint arthritis, carditis, and a rising antistreptolysin O titer about 3 weeks after untreated pharyngitis is acute rheumatic fever.

· The mechanism is molecular mimicry: the M protein is an alpha-helical coiled coil resembling cardiac myosin and tropomyosin, and antibody to the group A carbohydrate cross-reacts with valve laminin.

Distinctions that matter

· The organism is absent from the heart lesions, and the 2 to 3 week latent period is the time taken to build the adaptive response.

Common misconceptions

· Looking for infection in the valve. The valve lesion of rheumatic fever is sterile.

Where it is taught: Section C02 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C03-Q02

Board-style question: Purpura, casts, and a very low C4

Correct answer: A. IgM rheumatoid factor bound to IgG forming complexes that deposit in vessel walls and consume the classical pathway

Why it is right: Hepatitis C drives clonal B cells to produce IgM rheumatoid factor that binds IgG; these complexes precipitate below 37 degrees C (98.6 degrees F), deposit in vessel walls, and consume the classical pathway, which is why C4 is very low and the biopsy shows granular deposits.

B is wrong: Antibodies against myeloperoxidase cause pauci-immune injury with normal complement and almost no immunoglobulin deposition, which does not fit granular deposits with a very low C4.

C is wrong: Antibody against type IV collagen in the basement membrane gives linear rather than granular immunofluorescence, as in anti-GBM disease.

D is wrong: Dendritic cell activation of CD4 T cells with medial granulomas describes giant cell and Takayasu arteritis, which leave no deposits and affect large arteries rather than dermal venules and glomeruli.

A competent response contains

· Palpable purpura with red cell casts, a very low C4 with near normal C3, rheumatoid factor, and hepatitis C is mixed cryoglobulinemic vasculitis.

· Granular immunoglobulin and complement in dermal venules confirms an immune complex mechanism.

Distinctions that matter

· A very low C4 points to classical pathway consumption by IgM-IgG complexes, which separates this from IgA vasculitis where complement is normal.

Common misconceptions

· Treating purpura with casts as one disease. The complement level and the immunofluorescence pattern are what identify the mechanism.

Where it is taught: Section C03 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C05-Q03

Board-style question: Hypotension five months after transplantation

Correct answer: C. Donor-specific antibody binding capillary endothelium with classical pathway activation

Why it is right: Donor-specific anti-HLA antibody binds capillary endothelium and activates the classical pathway, leaving covalently bound C4d, and the widespread microvascular injury causes more hemodynamic compromise than cellular rejection of similar grade; indirect allorecognition provided the CD4 help for the antibody to develop after transplantation.

A is wrong: CD8 T cells recognizing intact donor HLA on passenger dendritic cells cause acute cellular rejection, which shows perivascular and interstitial lymphocytes with myocyte damage and no C4d.

B is wrong: PDGF and TGF-beta driven smooth muscle migration into the intima is cardiac allograft vasculopathy, which develops over months to years as diffuse concentric intimal thickening rather than acute capillary injury.

D is wrong: Preformed anti-ABO IgM causes hyperacute rejection within minutes to hours of reperfusion with small vessel thrombosis and necrosis, not an illness beginning 5 months later.

A competent response contains

· Capillary C4d with endothelial swelling, intravascular macrophages, few interstitial lymphocytes, and a newly detected donor-specific anti-HLA antibody is antibody-mediated rejection.

· C4d binds covalently where C4 was cleaved, so it marks tissue in which the classical pathway was activated.

Distinctions that matter

· Antibody-mediated rejection is microvascular and complement marked, while cellular rejection is interstitial and lymphocytic.

Common misconceptions

· Assuming antibody-mediated rejection must occur immediately. It occurs from days to years, once donor-specific antibody appears.

Where it is taught: Section C05 of the CV-16 Immune Responses in Cardiovascular Disease notes.

NPLEX-P1-CARD-16-01-C06-Q04

Board-style question: A young man two weeks after a viral illness

Correct answer: D. CD8 T cells killing infected myocytes with perforin and granzymes

Why it is right: From about 1 to 2 weeks CD8 T cells kill infected myocytes with perforin, granzymes, and Fas ligand while CD4 cells release IFN-gamma and TNF; this clears the virus but accounts for most of the muscle loss, and it is the phase this patient has reached.

A is wrong: Cleavage of dystrophin by viral protease 2A is the phase 1 mechanism acting over the first days, before the T-cell response has expanded.

B is wrong: Eosinophil major basic protein injures the myocardium in hypereosinophilic syndrome and eosinophilic granulomatosis with polyangiitis, which present with eosinophilia and endocardial disease rather than after a brief viral illness.

C is wrong: Occlusion of an epicardial coronary artery produces necrosis beginning subendocardially within one artery territory, whereas the angiogram is normal and the enhancement here is subepicardial and crosses territories.

A competent response contains

· A viral prodrome, a raised troponin, ST changes crossing coronary territories, normal arteries, and subepicardial late gadolinium enhancement is viral myocarditis.

· At about 12 days the patient is in the immune phase, which begins from about 1 to 2 weeks.

Distinctions that matter

· Direct viral killing dominates the first few days; the T-cell phase accounts for most myocyte loss from about 1 to 2 weeks.

Common misconceptions

· Attributing all myocyte loss in myocarditis to the virus itself. The response that clears it causes most of the damage.

Where it is taught: Section C06 of the CV-16 Immune Responses in Cardiovascular Disease notes.